
QXU5010

Surfaces and Interfaces

QMES ICT #4 – All Chapters – 2025/26

Question Book

Time allowed: 1 hour

Write ALL the answers in the Answer Book ONLY (Extra sheets not allowed)

Complete the information below about yourself very carefully.

QM student number

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NPU student number

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NOT allowed: electronic dictionaries

Multiple Choice Questions. For each question, choose the answer that you think is correct. There is only 1 correct answer to each question. Each correct answer is worth 1 mark. [15 marks]

- Q1. According to the module, what is a key distinction between the terms "surface" and "interface"?
- a) A surface is ALWAYS a two-dimensional, while an interface is ALWAYS three-dimensional
 - b) An interface is the preferred term to describe the boundary between two condensed phases**
 - c) A surface has higher energy than an interface
 - d) Surfaces only exist in solids, while interfaces only exist in liquids
- Q2. Why is a high surface-to-volume ratio particularly important in nanotechnology?
- a) It decreases the material's density, making it lighter
 - b) It makes gravitational forces dominant over intermolecular forces
 - c) It increases the area available for surface-mediated interactions and processes**
 - d) It ensures that the material behaves according to classical Newtonian mechanics
- Q3. What is the fundamental driving force behind the "hydrophobic effect" at the molecular level?
- a) The covalent bonding between water molecules and hydrophobic groups
 - b) The repulsion of hydrophobic groups by the hydrogen-bonded network of water**
 - c) The magnetic attraction between non-polar molecules
 - d) The increased solubility of hydrophobic groups in cold water
- Q4. In the context of liposomes, what is the primary functional advantage of using them as drug carriers?
- a) They permanently fuse with cell membranes, releasing all contents at once
 - b) They can solubilise hydrophobic drugs and protect their contents from metabolic enzymes**
 - c) They are naturally attracted to and destroy cancerous cells without any modification
 - d) They are the most cost-effective method for all drug delivery applications
- Q5. What is the primary conceptual difference between physisorption and chemisorption?
- a) Physisorption only occurs at high temperatures, while chemisorption occurs at low temperatures
 - b) Physisorption involves weak van der Waals forces, while chemisorption involves the formation of a chemical bond**
 - c) Physisorption is always irreversible, while chemisorption is always reversible
 - d) Physisorption increases surface energy, while chemisorption decreases it
- Q6. According to Bancroft's Rule, what primarily determines whether an Oil-in-Water (O/W) or Water-in-Oil (W/O) emulsion is formed?
- a) The relative volume of the oil and water phases
 - b) The temperature at which the emulsion is prepared
 - c) The phase in which the emulsifying agent is more soluble**
 - d) The viscosity of the continuous phase

Q7. What is a key characteristic of Self-Assembled Monolayers (SAMs) formed by alkanethiols on a gold surface?

- a) They are highly disorganised and form a liquid-like layer
- b) They are formed through physisorption and are easily washed off
- c) They form a densely packed, ordered array of molecules on the substrate**
- d) They require a complex, multi-step chemical synthesis to attach to the surface

Q8. For a thermally activated physical vapour deposition process, which of the following is NOT a requirement?

- a) Vacuum/low-pressure chamber
- b) Low reactivity of source material
- c) High reactivity of source material**
- d) A tuneable heating method (e.g. resistive heating)

Q9. Which of the following techniques can be correctly described as a top-down fabrication technique?

- a) Atomic Layer Deposition (ALD)
- b) Chemical Vapour Deposition (CVD)
- c) Sol-gel synthesis
- d) Ball milling**

Q10. Which of the following lithography techniques will enable the highest resolution removal of material from a surface?

- a) Optical lithography deposition
- b) Optical lithography etching
- c) Electron beam (e-beam) lithography deposition
- d) Electron beam (e-beam) lithography etching**

Q11. Which of the following wetting classifications can generally only be achieved through the introduction of surface roughness?

- a) Hydrophobicity
- b) Superhydrophobicity**
- c) Hydrophilicity
- d) Superhydrophilicity

Q12. Which of the following adhesive mechanisms, within an adhesive joint, relies upon an adhesive with a relatively high surface tension?

- a) Physical adhesion**
- b) Chemical adhesion
- c) Mechanical adhesion
- d) All of the above

Q13. Which of the following surfaces is generally expected to have the highest amount of bacterial attachment?

- a) **A highly rough material with a hydrophilic surface chemistry**
- b) A highly rough material with a hydrophobic surface chemistry
- c) A smooth material with a hydrophilic surface chemistry
- d) A smooth material with a hydrophobic surface chemistry

Q14. Which of the following is the most important to consider when using Scanning Electron Microscopy for imaging a material?

- a) The thickness of the sample
- b) The elemental composition of the sample
- c) The wetting properties of the sample
- d) **The electrical conductivity of the sample's surface**

Q15. Which of the following analysis techniques uses the energy released as electromagnetic radiation as the primary method of detection?

- a) **Energy Dispersive X-ray (EDX) Spectroscopy**
- b) Electron Diffraction (ED)
- c) X-ray Adsorption Spectroscopy (XAS)
- d) X-ray Photoelectron Spectroscopy (XPS)

*** Mark answers on this question/answer sheet [no extra sheets allowed]**

Question 1.**Answer ALL Parts of Question****[30 Marks]****Part (a)**

The formation and stability of liposomes are governed by the molecular properties of their constituent surfactants and the surrounding environment.

- i. State the two key phases of a surfactant bilayer and the critical temperature that governs the transition between them. What is the significance of this temperature for liposome formation? **[4 Marks]**

Answer:

Phases: (2 marks)

- Gel (solid-like) phase: hydrocarbon chains are ordered and tightly packed.
- Liquid-crystalline (fluid) phase: chains are disordered and mobile.

Critical temperature: (1 mark)

- Main phase transition temperature (T_m): the temperature at which the bilayer transitions from gel to liquid-crystalline phase.

Significance: (1 mark)

- At T_m , the membrane becomes fluid enough for liposome formation. Below T_m , the bilayer is rigid and cannot form stable vesicles.

- ii. The Critical Packing Parameter (CPP) is a useful concept for predicting molecular self-assembly. State the CPP value range that favours the formation of stable bilayers and liposomes. **[3 Marks]**

Answer:

- $CPP \approx 1$ (typically 0.5–1) favors bilayer formation and hence liposomes. (1.5 marks)
- $CPP < 1/3 \rightarrow$ micelles; $CPP > 1 \rightarrow$ inverted structures. (1.5 marks)

- iii. Liposomes can swell or shrink in response to the osmolarity of their environment. Explain what happens to a liposome placed in a hypotonic solution and the potential consequences for its membrane. **[4 Marks]**

Answer:

Hypotonic environment: (2 marks)

- External solution has lower osmolarity than inside the liposome.
- Water flows into the liposome by osmosis.

Effect: (2 marks)

- Liposome swells, increasing internal pressure.
- Potential consequence: Membrane may rupture if the tension exceeds its elastic limit.

- iv. Contrast 'Passive Transport' and 'Facilitated Transport' of a solute across a liposome membrane, focusing on the role of energy and membrane components. **[4 Marks]**

Answer:

Passive Transport: (2 marks)

- Movement of solute down its concentration gradient.
- No energy required.
- Occurs by simple diffusion through the lipid bilayer.

Facilitated Transport: (2 marks)

- Still down the gradient but requires specific membrane proteins (channels or carriers).
- No ATP but depends on protein-mediated pathways for selectivity and speed.

Part (b)

i. Name a coating deposition method that results in a flat surface morphology. [1 Mark]

Answer:

- ALD, CVD (Layer-by-layer), or other flat fabrication

ii. For your coating deposition method (in **1bi**), list the key steps in the deposition. Starting from precursors and ending with a coating. [5 Marks]

Answer:

- Precursor 'dosing' – controlled introduction.
- Physisorption onto substrate – 'self-limiting'.
- Purge; low pressure removes excess adsorbed precursor.
- Activator 'dosing', and surface reaction.
- Purge; excess adsorbed activator.

iii. For the deposition method described in **1bii**, identify the key steps that cause the coating to be flat, briefly explaining why this is the case. [4 Marks]

Answer:

- Physisorption is self-limiting.
- This limits multi-layer depositions.
- Purge steps remove incomplete layers.
- This leaves atomically flat coatings.

iv. A coating made using the above technique (**1bii**) has a water contact angle of 60°. When this is roughened by abrasion, the observed water contact angle decreases to 30°. Use the most suitable wetting model to calculate the degree of surface roughness made on the surface via abrasion. [5 Marks]

Answer:

- Wenzel Equation
- Insert $\theta = 60^\circ$, and $\theta_w = 30^\circ$
- Rearrange for r
- $R = 1.73$ [2 marks]